ACR-Net: Adaptive Correlation Refined Hyperspectral Unmixing

Meiqi Hu, *Member, IEEE*, Chen Wu, *Member, IEEE*, Rong Liu, *Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Hyperspectral unmixing (HU) aims to resolve the prevalent issue of mixed pixels in hyperspectral imagery and serves as an effective technique for subpixel-level image interpretation. Recent years have seen the emergence of advanced unmixing algorithms that integrate both spatial information and spectral information. However, existing methods mainly focus on spatial context and lack depth in modeling spatial correlations. Both relevant spatial information and irrelevant spatial information are introduced into the spectral mixing model for local pixels, with the irrelevant information acting as noise that impacts the unmixing accuracy. To address these challenges, we propose an advanced spectral mixing model, adaptive correlation refined network (ACR-Net) that integrates refined spatial correlation based on self-attention. A key component of ACR-Net is the adaptive correlation aggregated decoder (ACAD), which extracts affinity information from the encoder's feature map and adaptively amplifies the influence of highly correlated regions in the unmixing process. Additionally, the composite active spatial attention (CASA) module emphasizes unique spectral characteristics across bands, improving spatial distribution representation and enabling more accurate abundance estimation. We conducted abundant evaluations on six datasets, including three real-world and two synthetic HU datasets as well as a large benchmark classification dataset. Extensive experiments have demonstrated that the proposed algorithm can achieve exceptional or comparable unmixing results among various state-of-the-art algorithms. Moreover, the proposed method achieved optimal classification performance on the classic PaviaU dataset, indicating its strong potential for hyperspectral classification task.**

***Index Terms*— Adaptive correlation, hyperspectral unmixing (HU), linear unmixing model, spatial affinity.**

I. Introduction

THE fast development of hyperspectral imaging technology has provided a wider exploration space for fine-grained hyperspectral processing tasks [1], [2]. Owing to the detailed spectral information and image-spectrum merging characteristic [3], hyperspectral images (HSIs) gain extensive application in precise classification [4], target detection [5], change detection [6], and spectral unmixing [7]. Under the constraint of limited spatial resolution [8], the ground objects in HSIs usually appear in mixed pixels, increasing the difficulty of precise object recognition and extraction.

Hyperspectral unmixing (HU) [9] targets at decomposing the mixed pixel into a combination of pure materials (called endmember) and corresponding proportions (called abundance), serving as a significant preprocessing step for high-level data analysis. Typical applications of HU include mineral exploration [10], heavy metal pollution detection in ecological environmental monitoring [11], image fusion and synthesis [12], and so on.

A. Classic Unmixing

HU can be either based on the linear mixing model (LMM) [13] or the nonlinear mixing model (NLMM) [14], [15]. The complexity of the NLMM model is raised by the need of incorporating prior knowledge about the scene and the materials being examined for NLMM [16], [17]. By contrast, LMM is computationally effective and easier to understand. This simplicity and efficiency have contributed to the widespread adoption of LMM as the dominant approach.

HU algorithms grounded in LMM can be categorized into two primary groups according to their procedural structure. The first group involves a dual-phase process: the extraction of endmembers, followed by abundance estimation [18]. Endmembers are either sourced from prior knowledge (spectral libraries containing endmembers of ground objects) [19], [20] or identified utilizing endmember extraction algorithms [21]. Spectral libraries are typically collected from the field or laboratory, making them challenging to acquire. Endmember extraction algorithms are commonly employed and can be primarily divided into three categories: geometric-based [22], [23], [24], spatial–spectral-based [25], [26], and intelligent evolutionary-based algorithms [27], [28]. These algorithms either extract a single endmember for each endmember class [23], [29] or extract a bundle of endmembers for each class, considering the spectral variability problem caused by different imaging conditions or material properties [28], [30]. Following endmember extraction, abundance estimation typically employs techniques such as constrained least squares (CLS) algorithms [31] or sparse unmixing algorithms [32], [33]. In contrast, the second group involves methods that simultaneously estimate both endmembers and

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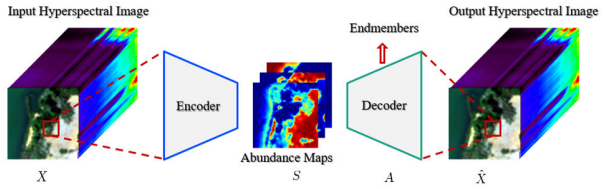


Fig. 1. Sketch map of AE-based spectral unmixing algorithm.

abundances [34], [35], [36]. Predominantly utilized methods within this category include those based on nonnegative matrix factorization (NMF) [37], [38] and Bayesian approaches [35], [39].

B. Deep Learning-Based Unmixing

In recent years, deep learning technology has demonstrated significant advantages in the field of HU, particularly through the integration of spatial–spectral information and various constraints in the learning process to achieve end-to-end unmixing [40], [41]. Many researchers have observed the strong connection between autoencoders (AEs) and unmixing, leading to the development of numerous AE-based unmixing algorithms [42], [43].

Generally, AE [44] consists of an encoder and a decoder. The encoder transforms high-dimensional hyperspectral information into a low-dimensional feature representation, while the decoder reconstructs it back to the original image space. The encoder compresses the input data, compelling the model to encapsulate the data within a constrained dimensional space. Consequently, the encoder must distill the most salient features of the data to enable the decoder to accurately reconstruct the original input [45]. As shown in Fig. 1, the process of unmixing can be well-matched with the structure of AE [46]. The decoding process can be written as $\hat{X} = AS$, where $\hat{X}$ refers to the reconstructed HSI, the weight of the decoder A corresponds to the endmember matrix exactly, and the low-dimensional representation obtained from the encoder is considered to be the abundance map S. Automatic unmixing can be achieved by minimizing the difference between the reconstructed spectrum and the original spectrum conveniently and regularization strategy.

Early unmixing works based on AE were mostly achieved by fully connected layers. The fully connected AE is capable of processing long dimensional spectral information, with the linear decoder weight exactly corresponding to the endmember matrix. Qu and Qi [42] proposed a novel untied denoising AE with sparsity to allow for decoupling between encoder and decoder. Deep autoencoder network (DAEN) [47] designed a deep stacked AE to learn a good endmember initialization from pure pixels, which was then fed into a tailored variational AE to perform hyperspectral unmixing. A novel self-supervised robust deep matrix factorization (SSRDMF) model that explicitly models sparse noise and includes a self-supervised constraint to explore spectral information is developed for HU [48]. Multitask autoencoder unxmixing (MTAEU) [49] adopts multiple parallel fully connected AEs to achieve simultaneous unmixing for a neighbor patch with a sharing hidden layer. Independent pixel unmixing works, however, find it difficult to combine the spatial structure and information necessary to create spatially related unmixing.

In order to alleviate the problem, convolutional neural network autoencoder unmixing (CNNAEU) [36] presented a spatial–spectral HU method based on convolutional neural network. Instead of convolutional layers, adversarial autoencoder network (AAENet) [50] cleverly introduced spatial distribution into the unmixing process through the use of adversarial training, under the assumption of homogeneous superpixels sharing the same distributions. A new sparsity-constrained deep NMF with total variation (SDNMF-TV) technique is designed for HU to exploit adequately the spectral and spatial information [51]. Cycle-consistency unmixing network (CyCU) [52] then proposed a circle cascaded AE network to keep cycle consistency between the original HSI and reconstructed HSI by the cascaded AE, preserving the high-level semantic information during the unmixing. DeepTrans [53] attempted to capture the global spatial feature by integrating convolutional AE and transformer. Joint metric neural network (JMnet) [54] introduced a discriminator to improve the distribution similarity between the reconstruction and observation by Wasserstein distance regularization terms. Endmember-guided unmixing network (EGU-Net) [40] employed the prior information from endmember to correct weights of the other Siamese unmixing network. Multiview spatial-spectral two-stream network (MSSS-Net) [41] introduced a novel two-stream AE for enhanced spatial–spectral unmixing by leveraging multiview spectral and spatial data. In addition, bilinear hyperspectral unmixing network (BU-Net) proposes an additional AE-based stream for selected spectral channels with interclass separability to enhance spectral discriminability [55].

In summary, numerous advanced deep learning-based HU algorithms have emerged, enhancing unmixing performance by integrating spatial and spectral information and modeling spatial dependencies. However, a critical issue remains, where irrelevant features introduced through regular spatial information, can negatively impact endmember and abundance modeling, and superpixel-based spatial integration fails to capture precise spatial correlation degree. This lack of accurate spatial correlation modeling restricts the effective use of spatial continuity and similarity, thereby limiting the potential of spatial information to achieve accurate unmixing.

C. Motivation

Current approaches that integrate spatial information for HU primarily focus on incorporating spatial context but fall short in modeling spatial correlation in depth. However, the spatial relationships between target objects and surroundings provide valuable continuity and similarity information crucial for accurate unmixing. Without precise spatial correlation modeling, irrelevant or weakly correlated pixels may dilute the unmixing accuracy, limiting the effectiveness of endmember and abundance estimation.

Our goal is to address this gap by dynamically adjusting the specific spatial relationships of each pixel, enhancing the precision of endmember extraction and abundance estimation, ultimately leading to more accurate and reliable HU results.

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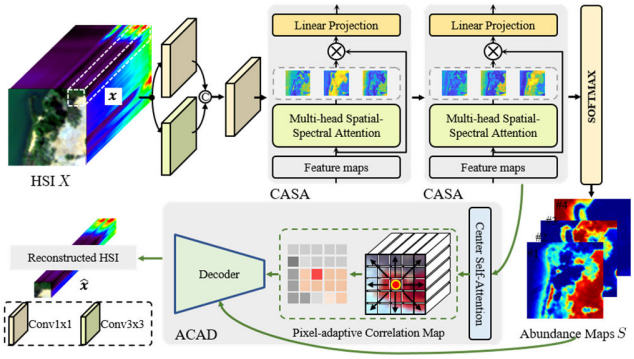


Fig. 2. Schematic of the proposed ACR-Net.

D. Contributions

A new adaptive correlation refined network (ACR-Net) is proposed for adaptive spatial relationship modeling (as shown in Fig. 2). Concretely, an adaptive correlation aggregated decoder (ACAD) is tailored to adaptively adjust the contribution of neighboring pixels, focusing on relevant regions to achieve precise unmixing. ACAD extracts pixel affinity from the encoder's spatial–spectral features to fine-tune contributions from neighbors, creating a more accurate spectral unmixing model. It is the first time integrating the self-attention mechanism into the modeling of combination of endmember and abundances, which transforms the conventional LMM into a refined spatial correlation model. Additionally, the designed composite active spatial attention (CASA) module leverages unique spectral characteristics across bands to improve feature discrimination.

The proposed adaptive spectral mixing model is capable of locally adjusting the combination of endmembers and abundances to account for globally varying nonlinear interactions, greatly enhancing the generalization. Abundant experiments on real and synthetic datasets confirm ACR-Net's effectiveness, demonstrating that CASA enhances abundance performance, while ACAD significantly improves endmember estimation accuracy.

The remainder of this article is presented as follows. Section II will give a detailed description of the proposed ACR-Net. The experimental results and analyses will be described in Section III. Finally, Section IV will summarize this article.

II. Methodology

The architecture of the proposed ACR-Net is presented in Fig. 2. The hyperspectral patch x is first processed by two different scaled convolutional layers to extract variant ground object features of different sizes. Since the salient features of different ground objects are shown in different channels, the CASA module is especially designed to activate and separate the discriminative features of various types of ground objects. The softmax operation is then employed to transform the extracted features into the abundance maps that follow the sum-to-one constraint. Afterward, we introduce an innovative ACAD to aggregate local neighborhood correlations to customize the reconstruction of the accurate central pixel. Neighborhood pixels with strong correlation contribute more to the reconstruction of the central pixel, while neighboring pixels with weak correlation make little contribution. The correlation exits in the extracted spatial–spectral feature maps and is modeled by the ACAD. Subsequently, ACAD establishes a precise unmixing model with pixel-related spatial information, removing the influence of irrelevant region to the unmixing process. Finally, the unmixing process is achieved by minimizing the loss between the reconstructed HSI and the original HSI and the constraint on the abundance maps. Note that the abundance map can be acquired from the encoder and the weight of decoder is served as an endmember matrix after training.

A. Proposed CASA Module

Essentially, unmixing involves decomposing individual mixed pixels into combinations of pure materials of different classes based on the spatial distribution proportions of objects. It becomes crucial to differentiate the regions of similar mixing combinations on a global scale in order to obtain accurate spatial distribution in the abundance result for each material.

Within the commonly utilized spectrum range for object imaging, spanning from visible light to near infrared, the distinctive attributes of various objects manifest across different channels due to the intricate spectral characteristics present in HSIs. As a result, it becomes feasible to distinguish between different categories of objects within the entire global image by articulating the notable features associated with different channels. The CASA module is tailored to emphasize different ground objects simultaneously, taking into account the discriminant characteristics of various ground objects in different bands on the spatial scale.

The details of CASA module are depicted in Fig. 3. Since the extracted feature maps inherently incorporate significant features of objects within certain bands, which are differentiated and displayed on different feature maps, it is sensible to group these feature maps into subspaces for highlighting and distinguishing different objects. Given the feature map gained from basic convolutional layers as $F \in \mathbb{R}^{h\times w\times b}$, where h, w, and b are the height, width, and channel of the feature map, respectively. We evenly divide F into n groups (same as the number of endmember), which can be denoted as $\{F_i \in \mathbb{R}^{h\times w\times (b/n)} | i = 1, 2, \ldots, n\}$, with each group containing (b/n) feature maps. Such a grouping strategy enables to focus on significant objects even less dominant within each group, rather than concentrating solely on the most prominent objects from all feature maps at once. Two descriptors max and mean operation are deployed to characterize the statistical features and object distribution within each group. Specifically, the max operation compresses a set of feature maps with (b/n) channels into one channel by calculating the maximum of the group feature maps, and the mean operation does the same by computing the mean value. Therefore, these two statistical feature maps $S_i^{\max}$ and S_i^{mean} can succinctly capture the noteworthy objects within the spatial domain of this group. The mathematical expressions can be summarized as follows:

$$S_i^{\max} = \max(F_i) \tag{1}$$

$$S_i^{\text{mean}} = \text{mean}(F_i) \tag{2}$$

$$S_i^{*} = \text{concat}\left(\left[S_i^{\max}, S_i^{\text{mean}}\right]\right). \tag{3}$$

Afterward, a learnable convolutional layer is deployed to map the two extracted features, enriching the statistical feature

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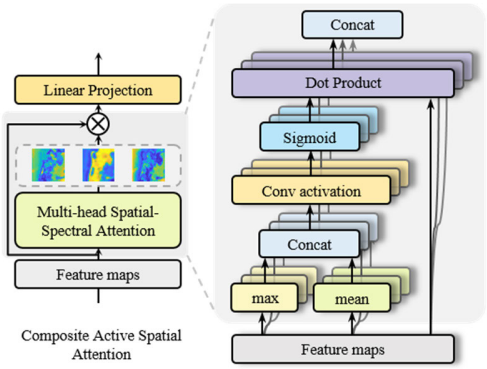


Fig. 3. Architecture of the proposed CASA module.

representation of this group's feature map and adaptively tracking the representation of salient features in the meantime. The learned statistical feature is then processed by the sigmoid activation function σ and is transformed into a weight map A_i with value ranging from 0 to 1. A higher value indicates a more pronounced representation of objects at that spatial position within the group

$$A_i = \sigma\left(\text{conv}_{7\times7}\left(S_i^*\right)\right). \tag{4}$$

A total of n spatial–spectral attention maps are obtained from the input feature map of the previous layer, and the weighted maps in different groups reflect prominent objects in spatial regions within different channel ranges. The multihead spatial–spectral attention maps $\{A_i|i = 1, 2, \ldots, n\}$ are then multiplied by the input feature map F to get an enhanced distinguished representation $\{H_i|i = 1, 2, \ldots, n\}$ that differentiates from certain ground objects to the other

$$H_i = A_i \cdot F_i. \tag{5}$$

Finally, the grouped activated features are concatenated and projected by a convolution layer with a kernel size of 1 to fuse variant grouped representations

$$O^{(j)} = \text{conv}_{1\times1}(\text{concat}([H_1, \ldots, H_n])) \tag{6}$$

where $O^{(j)}$ is the output of jth CASA module and the output of the final CASA module is processed through a softmax function to enforce an abundance map with a sum of one constraint.

The CASA module effectively utilizes the spectral features of various channels to activate various types of objects. Compared with widely used spatial attention and spectral attention, the group strategy generates diverse object attention maps, simultaneously focusing on various types of objects in space and effectively distinguishing between different objects. Multiple CASA modules facilitate the extraction of comprehensive spectral features from complex inputs. This is of great help to improve the abundance performance in unmixing tasks.

B. Proposed ACAD

Spatial context information from the neighborhood offers useful features about the background and spatial structure for center pixel unmixing. For the central pixel, each pixel in the local neighborhood contributes equally to its reconstruction for most AE-based unmixing algorithms. Actually, as shown in Fig. 4, different regions have different spatial distributions, and the correlation between central pixels and neighboring pixels also varies from pixel to pixel. Pixels corresponding to similar ground objects exhibit a strong correlation, whereas the correlation of pixels belonging to different ground objects is weak. When disregarding correlation and taking consideration of all neighboring pixels in equal contribution, the decoder weight, which serves as the endmember matrix, tends to be twisted due to the contribution of irrelevant object.

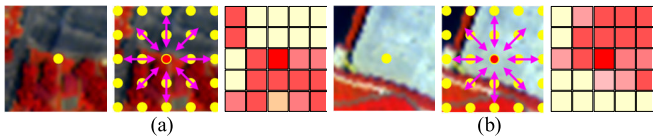


Fig. 4. (a) and (b) Two examples of center pixels which have different correlation with the local neighbor pixels. Pixels irrelevant to the center pixel have a negative impact on the reconstruction process of the center pixel.

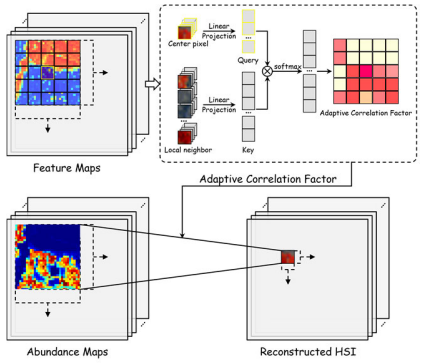


Fig. 5. Structure of ACAD.

To deal with the problem in unmixing model, ACAD is designed to adaptively model the correlation within the neighborhood, differentially describing the contribution of local neighboring pixels to the reconstruction of central pixels, achieving more accurate combination of endmember matrices and abundances. The details are presented in Fig. 5.

Mathematically, the modeling process of ACAD can be described as follows. For pixel p in an input patch, $x \in \mathbb{R}^{m\times m\times C}$, where m refers to the patch size and c means the number of channels of the input patch. Given the filter size as $k \times k$, the formulation that the convolutional decoder reconstructs the pixel p from the low-dimensional hidden representation S, which is the abundance map, is depicted as

$$\hat{x}_p = \sum_{i=1}^{k^2} \xi_i W_i s_i \tag{7}$$

where W_i is the weight matrix of decoder and ξ_i is the adaptive correlation factor. The above formula can be written as follows:

$$\hat{x}_p = \left(\sum_{i=1}^{k^2} \xi_i W_i\right) s_c + \sum_{i=1(i\neq c)}^{k^2} \xi_i W_i (s_i - s_c) \tag{8}$$

where s_c means the abundance of center pixel p and the second part represents the contribution value of the neighboring pixels to the reconstruction of the central pixel. ACAD accomplishes two vital functions: 1) each pixel in the neighborhood has a different correlation regulator ξ_i from the central pixel p and 2) the adaptive correlation factor ξ_i yields a positive correlation

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between the reconstruction contribution of different pixels to the central pixel and their affinity.

Noted that the correlation exits in the original HSI and also the extracted spatial–spectral feature maps. The correlation between two vectors can be described using the dot product. In particular, the center self-attention is designed to calculate the spatial correlation for each local neighbor. As mentioned above, the encoder extracts diverse spatial–spectral features, and the deeper feature maps gained from the CASA module characterize higher level semantic features, which are beneficial for distinguishing between different object and affinity calculations. Given the spatial–spectral feature as $\mathcal{H} \in \mathbb{R}^{m\times m\times r}$, where r is the number of channels of the feature, a certain pixel is denoted as $\mathcal{H}_i \in \mathbb{R}^{1\times 1\times r}$ and the pixels within a local neighborhood with a radius of $k(k < m)$ centered around the pixel $\mathcal{H}_i$ are referred to as $\mathcal{H}_i^N \in \mathbb{R}^{k\times k\times r}$. Linear projection is first employed to map the original feature vectors into embeddings, where the center pixel serves as the query $\mathcal{Q}_i \in \mathbb{R}^{1\times 1\times r}$, and the neighbor embeddings form the key $\mathcal{K}_i \in \mathbb{R}^{k\times k\times r}$. The correlation is calculated by performing dot product operations between the query and each pixel in the key. To ensure that the computed values do not become disproportionately large due to the high dimensionality of the vectors, we multiply the values obtained from the dot product calculation by a coefficient of $(1/\sqrt{r})$, where r represents the channel dimension of the query. Finally, the calculated correlation ξ_i of the local neighborhood is normalized to a range of 0–1 using softmax, where larger values reflect regions with strong correlation within the neighborhood range. This process can be summarized as follows:

$$\mathcal{Q}_i = M_q \mathcal{H}_i \tag{9}$$

$$\mathcal{K}_i = M_k \mathcal{H}_i^N \tag{10}$$

$$\xi_i = \text{softmax}\left(\frac{\mathcal{Q}_i \mathcal{K}_i}{\sqrt{r}}\right) \tag{11}$$

where M_q and M_k are projection matrices for generation of center embedding and neighbor embedding, respectively. The adaptive correlation factor ξ_i models the spatial correlation and structure, adaptively adjusting the contribution of neighbor pixels to the reconstruction of center pixel. The proposed ACAD provides a precise combination relationship of endmember and abundance maps with the help of spatial information, conducive to the accurate extraction of endmembers.

C. Loss Function

The loss function is composed of three parts. Spectral angle distance (SAD) is selected to enforce the coherence of the spectral reconstruction orientation. Notably, we apply an $L_{1/2}$ norm to the abundance, which serves as a moderate sparse constraint. Compared with L_0 constraint, $L_{1/2}$ norm yields an abundance vector where a dominant component occupies a significant portion instead of abundant zero abundance. This is based on the assumption that in HSIs, the proportion of components in the edge regions tends to be more uniform, while most pixels are located within a specific region, corresponding to a predominant component of a certain material. Also, a smooth regularization is posed on the endmember.

Given the original HSI as $x \in \mathbb{R}^{m\times m\times C}$ and the reconstructed HSI as $x \in \mathbb{R}^{m\times m\times C}$, where m is the patch size of input patch and C is the number of channel, for pixel i in the patch, the three mentioned loss functions are depicted as follows:

$$L_{\text{SAD}}(x_i, \hat{x}_i) = \arccos\left(\frac{x_i \hat{x}_i}{\|x_i\|_2 \left\|\hat{x}_i\right\|_2}\right) \tag{12}$$

$$L_{\text{smooth}}(E) = \sum_{i=1}^{R}\sum_{j=1}^{C-1}\left([E_i]_j - [E_i]_{j+1}\right)^2 \tag{13}$$

$$L_{1/2}(s_i) = \sum_{j=1}^{R} s_{ij}^{1/2} \tag{14}$$

where $E \in \mathbb{R}^{R\times C}$ is the endmember, R is the number of endmembers, and s_i is the abundance vector of pixel i in abundance map $s \in \mathbb{R}^{m\times m\times R}$.

The cost function of the proposed ACR-Net is defined as L, including three parts. Noted that α, β, and γ are coefficients for SAD loss, smooth loss, and $L_{1/2}$ norm, respectively, to keep the loss balance for better training

$$L = \alpha L_{\text{SAD}} + \beta L_{\text{smooth}} + \gamma L_{1/2}. \tag{15}$$

For all datasets, α and β are set as 1 and 0.5 according to the loss value, respectively. Since the abundance result is sensitive to the sparse constraint, a cosine decay strategy is adopted to adjust the coefficient of the sparse constraint during the training process as needed. The initial coefficient for $L_{1/2}$ norm is denoted as γ_{init}, and the end value of the coefficient is referred to as γ_{end}. For all datasets, γ_{init} is set as 0.5, while γ_{end} is set as 0.0. The discussion about the setting of γ is depicted in Section III-F.

After the training process, the endmembers are derived by computing the mean of the decoder's weights. On the other hand, the abundance map is obtained from the output of the encoder, which captures the spatial distribution of the material abundances within the given hyperspectral scene.

D. Overview of Network Architecture

To intuitively understand the network structure of the proposed method, Table I summarizes the network structure.

III. Experiments and Analysis

A. Dataset Description

To fully evaluate the performance of the proposed method, three real-world HU datasets and two synthesized datasets are opted for a test. In addition, another hyperspectral classification dataset has been used to testify the unmixing performance on the hyperspectral classification task.

1) Samson Dataset [56]: As shown in Fig. 6(a), the Samson dataset consists of a 95 × 95-pixel region with 156 spectral bands spanning wavelengths from 401 to 889 nm. The spectral resolution is as high as 3.13 nm. This dataset is free of blank or heavily noisy channels and contains three distinct target materials: 1) soil; 2) tree; and 3) water.

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TABLE I
CONFIGURATION OF THE PROPOSED ACR-NET'S NETWORK STRUCTURE

Network	Layer	Dim_in	Dim_out
Encoder	Conv2d, kernel=1; LeakyReLU(0.02); BatchNorm2d;	C	R*16
	Conv2d, kernel=3, pad=1; LeakyReLU(0.02); BatchNorm2d;	C	R*16
	Conv2d, kernel=1; LeakyReLU(0.02); BatchNorm2d;	R*32	R*16
	AvgPool1d; MaxPool1d; Conv2d, kernel=7, pad=3;	R*16	R*16
	Conv2d, kernel=1; LeakyReLU(0.02); BatchNorm2d;	R*32	R*8
	AvgPool1d; MaxPool1d; Conv2d, kernel=7, pad=3;	R*8	R*8
	Conv2d, kernel=1; LeakyReLU(0.02); BatchNorm2d;	R*8	R
	Softmax;	R	R
Decoder	Conv2d, kernel=1; Conv2d, kernel=1; Conv2d, kernel=5; ReLU();	R	C

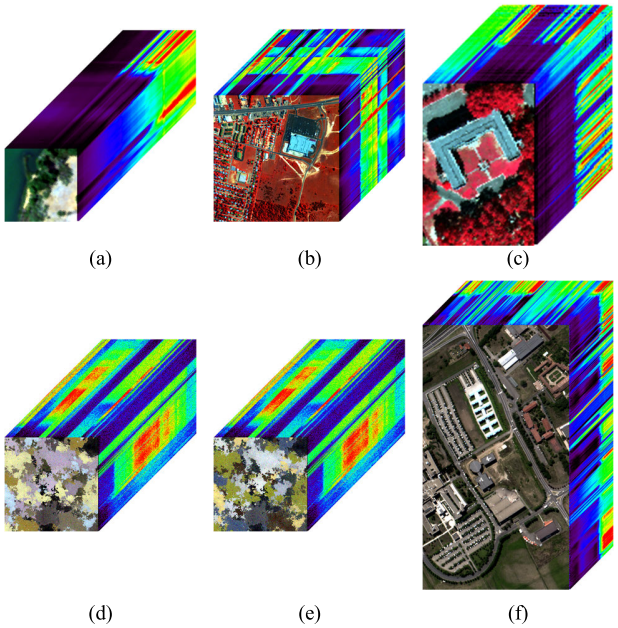


Fig. 6. Three-dimensional hyperspectral cube for four datasets. (a) Samson dataset, (b) Urban dataset, (c) LMM dataset, (d) Synthesized dataset: LMM ($R = 10$) (20 dB), (e) Synthesized dataset: PPNMM ($R = 10$) (20 dB), and (f) PaviaU dataset.

2) Urban Dataset [57]: As shown in Fig. 6(b), the Urban dataset owns a spatial size of 307 × 307 pixels, with 162 bands spanning from 400 to 2500 nm. Owing to the dense water vapor and atmospheric influences, channels 1–4, 76, 87, 101–111, 136–153, and 198–210 are removed from the original 210 channels. The spectral resolution is as high as 10 nm. There are four endmembers in this dataset, namely: 1) asphalt; 2) grass; 3) tree; and 4) roof.

3) Muffle Dataset [58]: This dataset was collected over the campus of Southern Mississippi Gulfpark. A popular region of interest with a size of 130 × 90 pixels was extracted, covering a spectral range from 0.375 to 1.050 μm, as illustrated in Fig. 6(c). Five predominant materials in this scene are investigated: 1) roof; 2) grass; 3) tree; 4) shadow; and 5) asphalt.

4) Synthesized Dataset: To thoroughly evaluate unmixing performance under controlled conditions, synthetic datasets are included in the analysis. Two synthetic datasets of size 256 × 256 are simulated with a number of endmembers set as 10 and signal-to-noise ratios (SNRs) as 20 dB. $R = 10$ endmembers are chosen from the United States Geological Survey (USGS) digital spectral library. A total of 269 bands are acquired by means of uniformly spaced sampling applied to the original thousands of bands. The abundance maps are generated using the hyperspectral imagery synthesis (HYDRA) toolbox.[1] The synthetic images were generated according to two distinct mixing mechanisms: LMM and polynomial post-nonlinear mixing model (PPNMM), as shown in Fig. 6(d) and (e).

5) PaviaU Dataset[2]: The Pavia University dataset, captured by the reflective optics system imaging spectrometer (ROSIS) sensor over Pavia, Italy, consists of 103 spectral bands spanning wavelengths between 0.43 and 0.86 μm. With a spatial resolution of 610 × 340 pixels and a ground resolution of roughly 1.3 m, this dataset includes nine labeled categories representing a variety of human-made and natural elements. Given the spectral similarity between bricks and gravel, as well as asphalt and bitumen, these were merged into joint categories. In our experiment, we focused on seven primary classes: 1) asphalt-bitumen; 2) meadows; 3) gravel bricks; 4) trees; 5) metal sheets; 6) bare soil; and 7) shadow.

B. Experimental Setup

1) Experimental Setting: The method is implemented by Pytorch and all experiments are processed by a single NVIDIA RTX 3090 GPU. We initialize the network by He-normal way [60] and use RMSprop as the optimizer with a weight decay of 0.001. For each dataset, 800 patches with a patch size of 13 are randomly selected from the original image for training. We set the batch size as 16 and epochs as 200 for all unmixing datasets except the PaviaU dataset. Given that the PaviaU dataset is large scale and contains a considerable number of ground objects, we set the number of epochs to 400 and the batch size to 32.

Regarding the complexity of each dataset, we roughly categorize all the datasets in this article into three types based on the data size and the quantity of pure substances. Type 1 features a small data size and a low quantity of pure substances, including the Samson, Urban, and Muffle datasets. Type 2 is characterized by a small data size yet a high quantity of pure substances, covering two simulated datasets. Type 3 has a large data size and a large quantity of pure substances, represented by the PaviaU dataset. Correspondingly, the training models for these three types of data are configured with two-, four-, and six-layer CASA modules. Other datasets can also take this criterion for reference as well. As for the learning rate, different values are adopted for the encoder and the decoder.

[1]https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Imagery_Synthesis_tools_for_MATLAB

[2]https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University

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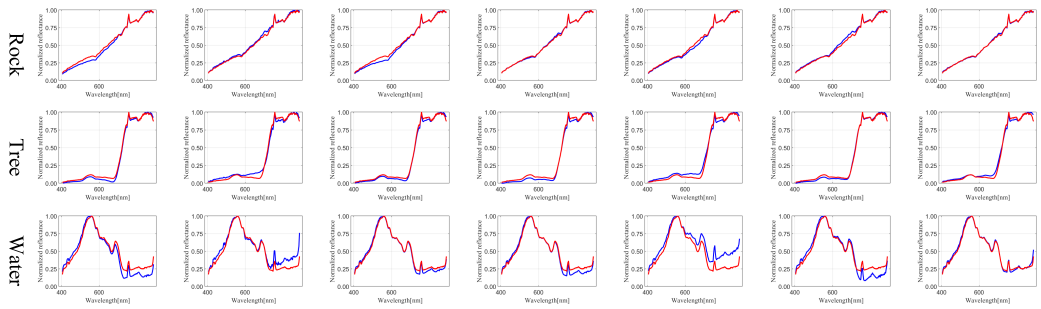


Fig. 7. Endmember prediction results of Samson dataset (references are in red, while the predictions are in blue). (a) ELMM, (b) CNNAEU, (c) MSNet, (d) DeepTrans, (e) GLU-OH, (f) MLM-3DAE, and (g) proposed ACR-Net.

TABLE II
HYPERPARAMETERS SETTING FOR ALL DATASETS

Hyperparameters	N_L	Epoch	Batch size	L_D
Samson	2	200	16	1e-4
Urban	2	200	16	1e-3
Muffle	2	200	16	5e-4
Synthesized Data	4	200	16	1e-5
PaviaU	6	400	32	1e-3

The learning rate of the encoder is fixed at 0.001, while the setting of decoder's learning rate L_D is shown in Table II. N_L refers to the number of layers of the CASA module.

2) Comparative Methods: Six advanced unmixing approaches are opted for comparison, including extended LMM (ELMM) [61], CNNAEU [36], multi-stage convolutional autoencoder network (MSNet) [62], DeepTrans [53], graph-based graph learning unmixing network based on one-hot pseudo-label (GLU-OH) [63], and multilinear mixing model based on 3-D convolutional layers (MLM-3DAE) [64]. Among them, CNNAEU, DeepTrans, MSNet, and GLU-OH are mainly based on the LMM. In contrast, ELMM employs the extended LMM, and MLM-3DAE adopts a multilinear mixing model. In terms of model architecture, CNNAEU, DeepTrans, MSNet, and MLM-3DAE are deep learning-based methods developed in recent years. GLU-OH applies graph-based learning to the unmixing problem.

3) Evaluation Index: To assess the performance of the proposed method, two mostly used quantitative evaluation indices are introduced. The average SAD (mSAD) is utilized to compare the similarity between the proposed endmembers and reference endmembers. It represents the angle between two high-dimensional vectors, and the closer this value is to 0, the more accurate the proposed endmembers are

$$L_{\mathrm{mSAD}}\left(m_i, \hat{m}_i\right) = \frac{1}{p}\sum_{i=1}^{p} \arccos\left(\frac{m_i \hat{m}_i^{\mathrm{T}}}{\|m_i\|_2 \left\|\hat{m}_i\right\|_2}\right) \tag{16}$$

where m_i and $\hat{m}_i$ are the extracted endmember and the reference endmember, respectively.

Additionally, we also calculate the mean squared error (MSE) value between each abundance map and its reference abundance. A smaller MSE value indicates smaller errors in the extracted abundances and better algorithm performance

$$L_{\mathrm{MSE}}\left(s_i, \hat{s}_i\right) = \frac{1}{p}\sum_{i=1}^{p}\left(s_i - \hat{s}_i\right)^2 \tag{17}$$

where s_i and $\hat{s}_i$ are the extracted and the reference abundance, respectively.

C. Experiments of Real Datasets

Tables III and IV provide quantitative evaluations on three real hyperspectral datasets, with optimal results highlighted in bold red and second-best results in bold italics.

1) Samson Dataset: The Samson dataset presents a relatively simple scene consisting of only three endmembers. Based on the quantitative results, several algorithms demonstrate strong unmixing performance on this dataset, with the proposed method achieving the highest overall accuracy (OA). As shown in Figs. 7 and 8, the visualization results further indicate that MSNet, DeepTrans, and ACR-Net gain effective performance in endmember predictions, with MSNet, GLU-OH, and ACR-Net excelling in precise endmember estimation.

2) Urban Dataset: The Urban dataset features a complex distribution of ground objects and contains a substantial number of mixed pixels. Despite these complexities, the proposed method attains the best performance in endmember extraction and ranks second in abundance prediction. The quantitative results of endmember and abundance are provided in Figs. 9 and 10. It is observed that ACR-Net, CNNAEU, and GLU-OH demonstrate high accuracy in endmember prediction, whereas ELMM and MLM-3DAE struggle with the accurate extraction of grass and roof endmembers for the Urban dataset, thereby limiting their unmixing performance.

3) Muffle Dataset: Captured by a drone, the Muffle dataset is similar in scene composition to the Urban dataset but has a narrower spectral range of 0.375–1.050 μm, providing less spectral information. The high spectral similarity between grass and tree, as well as roof and asphalt, further complicates endmember extraction. According to Tables III and IV, ACR-Net achieves the highest accuracy in unmixing performance, with DeepTrans and MSNet following second-best endmember extraction performance. Figs. 11 and 12 reveal that GLU-OH performs suboptimally, as its reliance on one-hot (OH) pseudo-labels for graph learning supervision may hinder initial graph construction when endmembers have high similarity. Overall, ACR-Net is capable of effectively distinguishing between different endmembers and provides precise abundance estimation, underscoring the efficacy of the proposed method.

D. Experiments of Synthesized Datasets

Table V summarizes the unmixing results on two synthetic datasets. MLM-3DAE ranks first for both synthetic datasets.

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TABLE III
SAD COMPARISON OF ABUNDANCE RESULTS FOR THREE REAL DATASETS

Dataset	Method	ELMM	CNNAEU	MSNet	DeepTrans	GLU-OH	MLM-3DAE	ACR-Net
Samson	Rock	0.1169	0.0281	0.0506	***0.0174***	0.0314	0.0265	**0.0125**
	Tree	0.0488	0.0677	**0.0365**	0.0532	0.0714	***0.0368***	0.0526
	Water	0.1283	0.1239	***0.0411***	0.0702	0.1664	0.1286	**0.0353**
	Average	0.0980	0.0732	***0.0427***	0.0469	0.0898	0.0639	**0.0335**
Urban	Asphalt	0.2283	***0.0809***	0.1291	0.0820	0.1124	0.1376	**0.0776**
	Grass	0.3929	**0.0372**	0.1488	0.3001	0.0550	1.2637	***0.0433***
	Tree	0.3045	0.1482	0.1491	***0.0863***	0.1280	0.1493	**0.0765**
	Roof	0.8237	***0.0795***	0.1564	0.1362	**0.0662**	0.1896	0.1267
	Average	0.4373	***0.0865***	0.1458	0.1512	0.0904	0.4351	**0.0810**
Muffle	Roof	0.1736	0.0455	**0.0153**	0.0675	0.2978	0.0709	***0.0341***
	Grass	0.1502	0.1078	0.1270	**0.0731**	0.0909	0.1346	***0.0738***
	Tree	0.1153	**0.0461**	***0.0577***	0.0700	0.1610	0.1393	0.0767
	Shadow	***0.1016***	0.1241	0.1073	**0.0662**	0.2170	0.1368	0.1019
	Asphalt	0.1883	0.1740	0.2006	0.1880	0.1696	***0.1614***	**0.1108**
	Average	0.1458	0.0995	0.1016	***0.0930***	0.1873	0.1286	**0.0795**

TABLE IV
MSE COMPARISON OF ABUNDANCE RESULTS FOR THREE REAL DATASETS

Dataset	Method	ELMM	CNNAEU	MSNet	DeepTrans	GLU-OH	MLM-3DAE	ACR-Net
Samson	Rock	0.0501	0.0582	***0.0081***	0.0227	0.0096	0.0392	**0.0032**
	Tree	0.0439	0.0369	***0.0042***	0.0199	0.0057	0.0478	**0.0011**
	Water	0.0123	0.0277	**0.0020**	0.0540	0.0041	0.1024	***0.0034***
	Average	0.0266	0.0409	***0.0047***	0.0322	0.0065	0.0632	**0.0025**
Urban	Asphalt	0.0211	0.0307	***0.0157***	**0.0138**	0.0207	0.0519	0.0193
	Grass	0.0573	0.0286	0.0386	0.0872	***0.0157***	0.1730	**0.0142**
	Tree	0.0657	0.0148	0.0154	0.0860	**0.0034**	0.0523	***0.0112***
	Roof	0.0229	0.0126	0.0154	***0.0083***	**0.0044**	0.0267	0.0147
	Average	0.0334	0.0216	0.0213	0.0488	**0.0110**	0.0760	***0.0149***
Muffle	Roof	***0.0210***	0.0493	**0.0183**	0.0409	0.0821	0.0484	0.0278
	Grass	0.1974	0.1170	0.1280	0.1181	0.1228	**0.0953**	***0.1070***
	Tree	0.0863	0.0600	0.0689	0.0586	0.1170	***0.0552***	**0.0505**
	Shadow	0.0745	0.0851	**0.0412**	0.0766	0.1770	0.0662	***0.0486***
	Asphalt	**0.0420**	0.0871	0.0520	0.0714	0.0672	***0.0504***	0.0602
	Average	0.0702	0.0797	***0.0617***	0.0731	0.1132	0.0631	**0.0588**

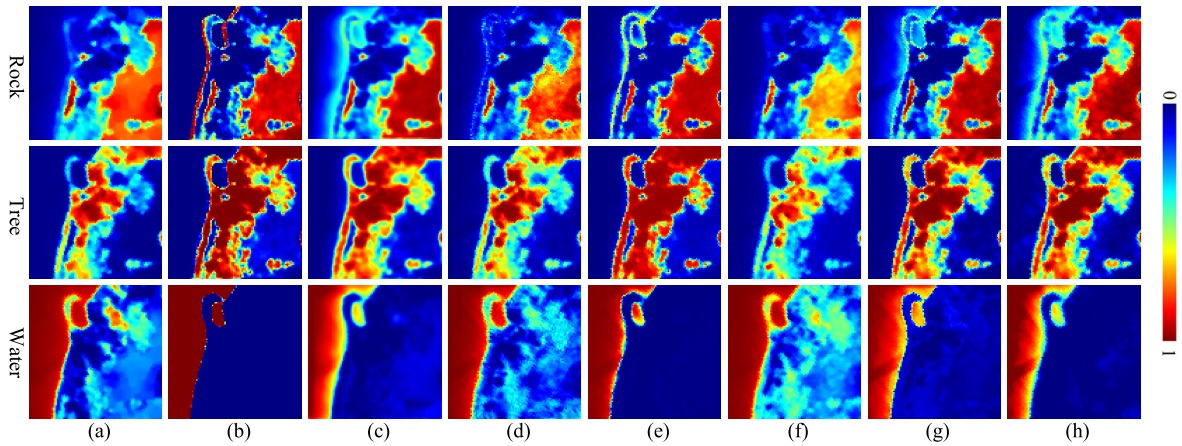


Fig. 8. Abundance map results of Samson dataset. (a) ELMM, (b) CNNAEU, (c) MSNet, (d) DeepTrans, (e) GLU-OH, (f) MLM-3DAE, (g) proposed ACR-Net, and (h) reference abundance map.

ACR-Net acquires comparable results under the complex mixing mode and with high noise level, indicating great generalization. Compared with the LMM, the extended LMM, and multilinear mixing model, the proposed adaptive spectral mixing model captures the spatial correlations between pixels and dynamically adjusts the contribution weights of neighboring pixels, enabling strong applicability in handling unmixing tasks in complex scenarios.

Fig. 13 visualizes the abundance prediction results on the challenging PPNMM-generated data at SNR $=$ 20 dB and $R = 10$. Here, MLM-3DAE and ACR-Net accurately predicted the proportions of each endmember in mixed pixels, closely aligning with reference results. Despite complex object distributions and noise interference, the proposed method maintained robust unmixing performance. Analysis shows that ACR-Net's weighted LMM uniquely captures spatial

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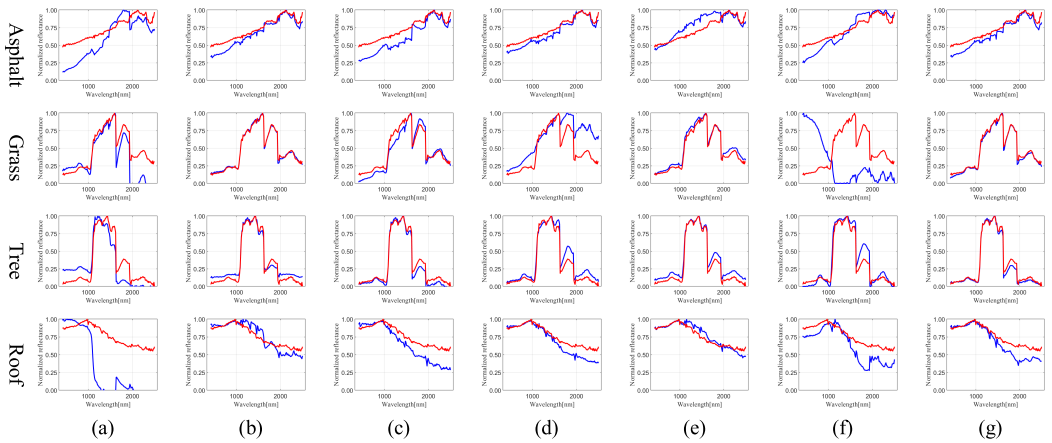


Fig. 9. Endmember prediction results of Urban dataset (references are in red, while the predictions are in blue). (a) ELMM, (b) CNNAEU, (c) MSNet, (d) DeepTrans, (e) GLU-OH, (f) MLM-3DAE, and (g) proposed ACR-Net.

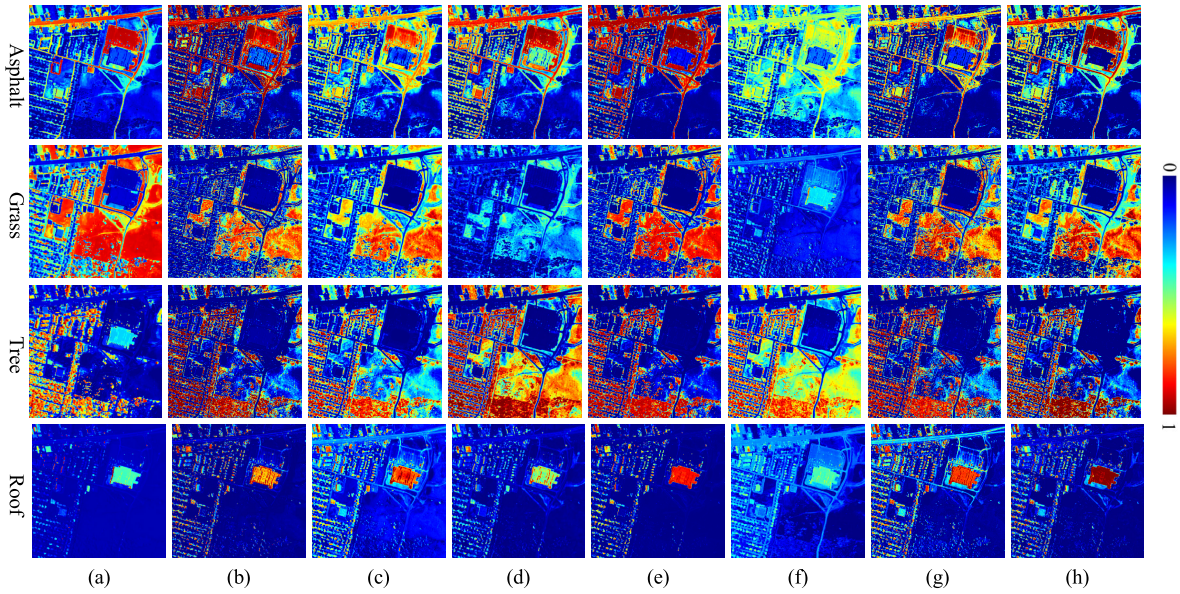


Fig. 10. Abundance map results of Urban dataset. (a) ELMM, (b) CNNAEU, (c) MSNet, (d) DeepTrans, (e) GLU-OH, (f) MLM-3DAE, (g) proposed ACR-Net, and (h) reference abundance map.

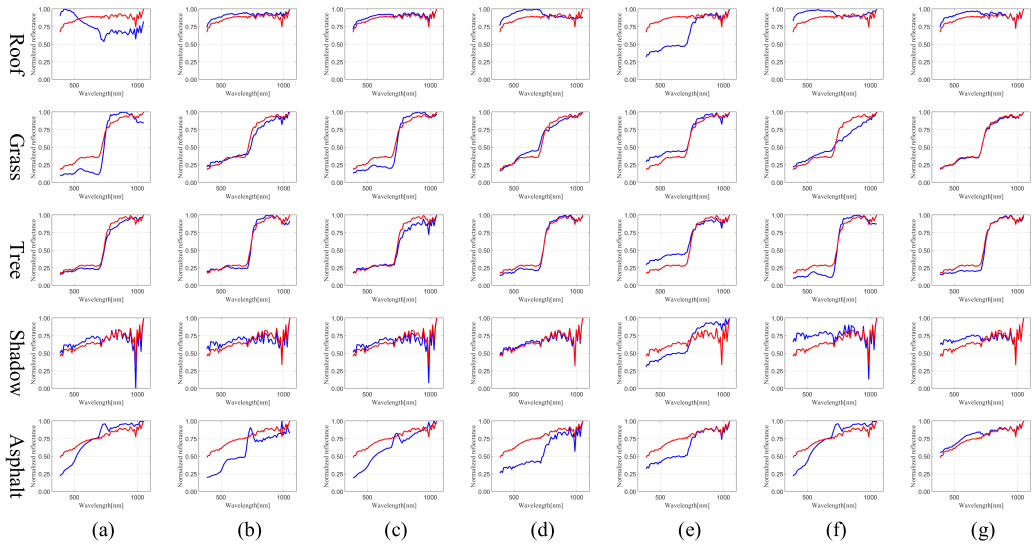


Fig. 11. Endmember prediction results of Muffle dataset (references are in red, while the predictions are in blue). (a) ELMM, (b) CNNAEU, (c) MSNet, (d) DeepTrans, (e) GLU-OH, (f) MLM-3DAE, and (g) proposed ACR-Net.

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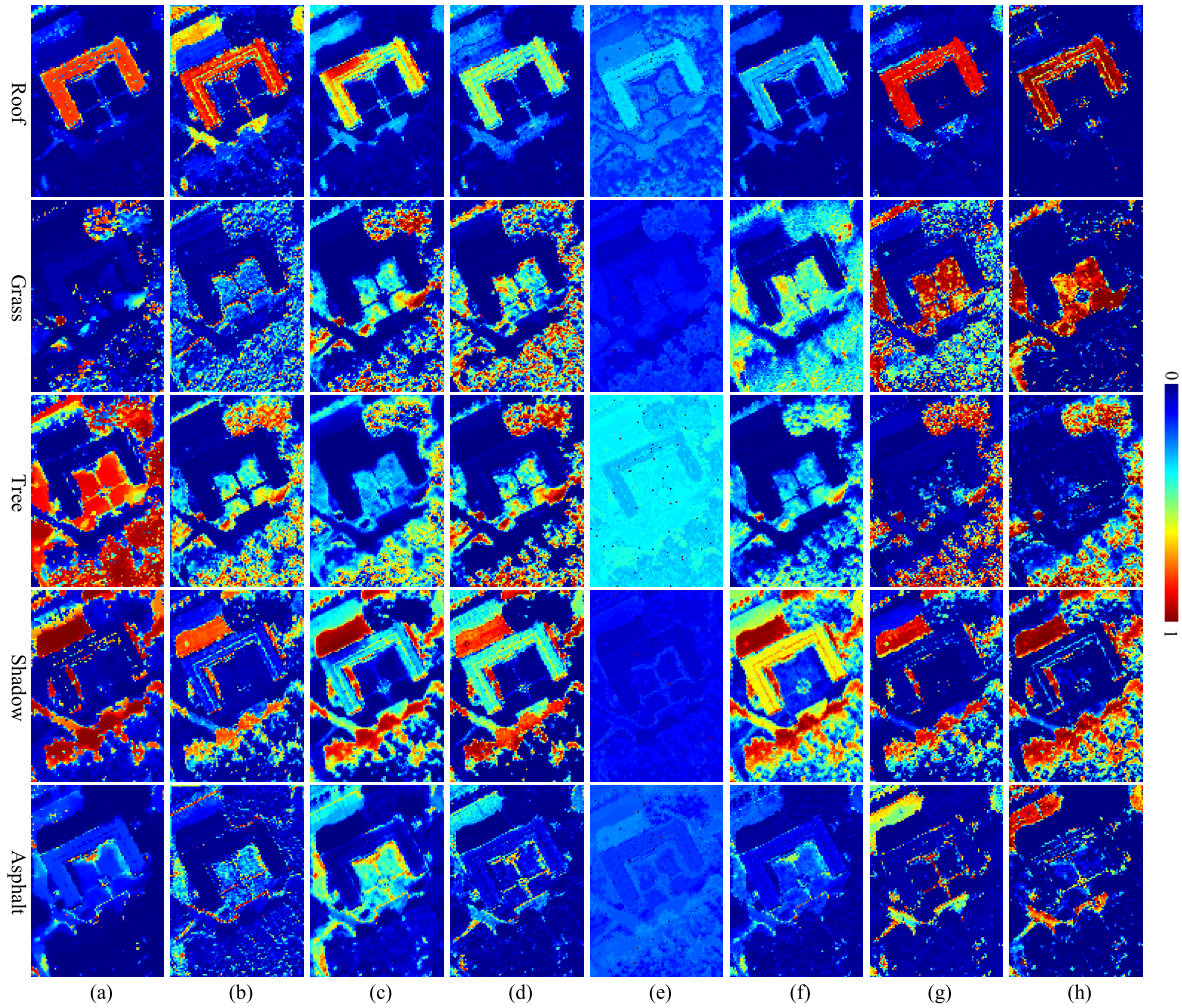


Fig. 12. Abundance map results of Muffle dataset. (a) ELMM, (b) CNNAEU, (c) MSNet, (d) DeepTrans, (e) GLU-OH, (f) MLM-3DAE, (g) proposed ACR-Net, and (h) reference abundance map.

TABLE V
COMPARISON OF UNMIXING RESULTS FOR SYNTHETIC DATASETS

	Dataset | Method	ELMM	CNNAEU	MSNet	DeepTrans	GLU-OH	MLM-3DAE	ACR-Net
SAD	LMM (R =10) (20dB)	0.0452	0.1179	0.0845	0.0659	0.1738	**0.0230**	***0.0441***
	PPNMM (R =10) (20dB)	***0.0439***	0.1504	0.0862	0.0681	0.1703	**0.0247**	0.0467
MSE	LMM (R =10) (20dB)	0.0118	0.0776	0.0127	0.0243	0.0741	**0.0039**	***0.0077***
	PPNMM (R =10) (20dB)	0.0124	0.1284	0.0150	0.0345	0.1306	**0.0036**	***0.0087***

correlations with neighboring pixels, offering a refined representation of spatial distributions.

E. Experiments of PaviaU Datasets

Table VI lists the assessment of classification result for each algorithm on PaviaU, including producer accuracy for each class, as well as OA and Kappa score as comprehensive evaluation metrics. Note that the land class with the highest abundance is assigned the final category. Due to the large image size, the computational resources required by GLU-OH are too high to run. The results show that ACR-Net achieves the best classification performance without labeled data, attaining an OA of 0.6740 and a Kappa score of 0.5736. MSNet ranks second. Fig. 14 visualizes the classification results, showing that most algorithms perform well in classifying metal sheets and gravel bricks. However, aside from the proposed method, most algorithms struggle to achieve accurate classification results of asphalt-bitumen. The results demonstrate that ACR-Net achieves outstanding abundance result even under complicated scenarios, verifying its effectiveness and highlighting the potential of unmixing in hyperspectral classification.

F. Discussion

1) Influence of Network Initialization: The experimental setup is detailed in Table VII. Three commonly used initialization methods, uniform, normal, and He-normal, are tested, along with a comparison of vertex component analysis (VCA)-based initialization for the endmembers. Four representative datasets are selected, including Muffle, LMM-generated data at SNR = 20 dB with $R = 10$ [LMM ($R = 10$) (20 dB)], PPNMM-generated data at SNR = 20 dB with $R = 10$

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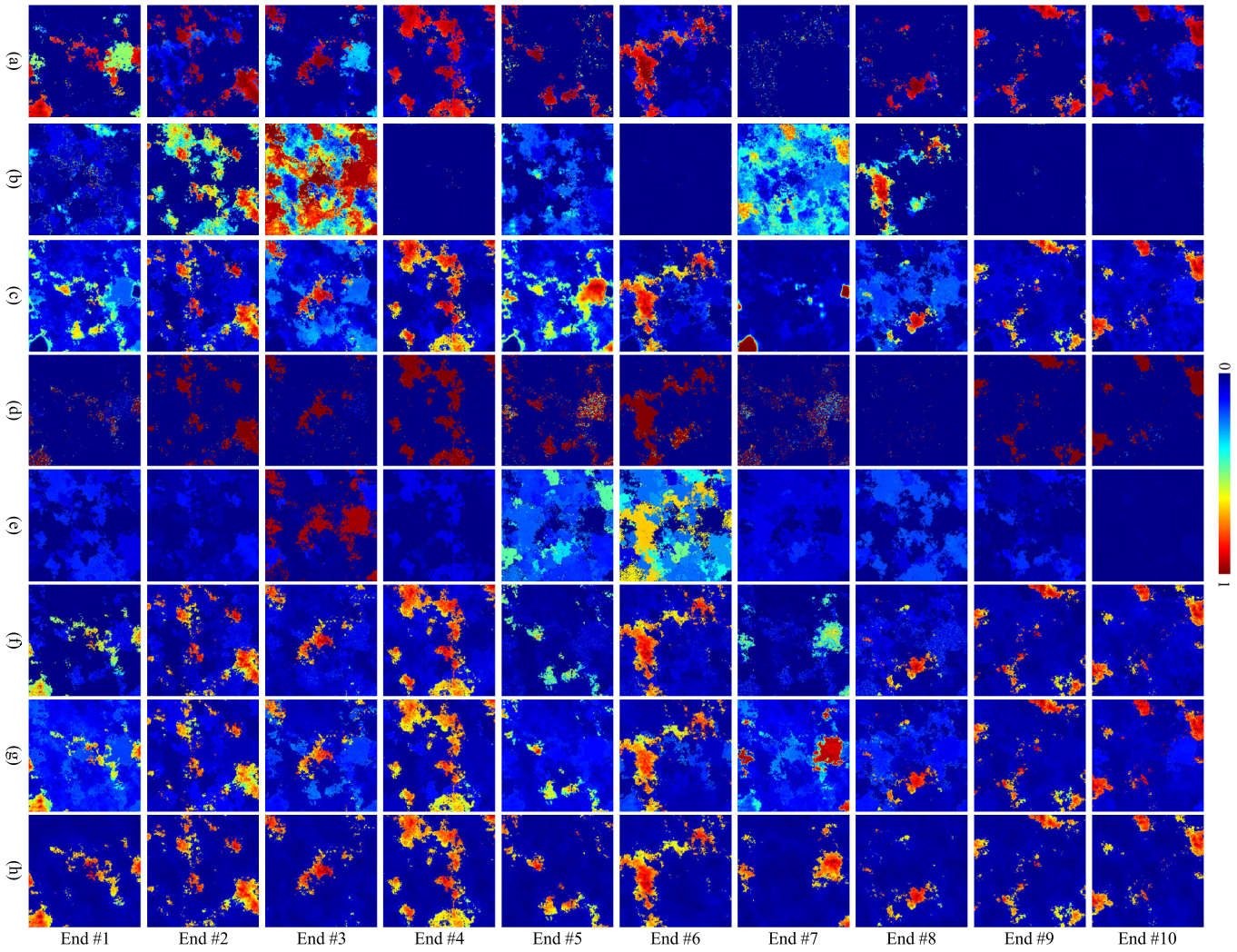


Fig. 13. Abundance map results on the PPNMM-generated data at SNR = 20 dB and $R = 10$. (a) ELMM, (b) CNNAEU, (c) MSNet, (d) DeepTrans, (e) GLU-OH, (f) MLM-3DAE, (g) proposed ACR-Net, and (h) reference abundance map.

TABLE VI

CLASSIFICATION COMPARISON FOR PAVIAU DATASET

Class	ELMM	CNNAEU	MSNet	DeepTrans	MLM-3DAE	ACR-Net
1	0.0000	0.0001	0.0000	***0.0008***	0.0000	**0.7829**
2	0.0013	0.5034	0.7235	**0.8854**	***0.8676***	0.6670
3	**0.9952**	0.9825	0.9922	***0.9948***	0.1296	0.7210
4	**0.9912**	0.7213	***0.9820***	0.0000	0.9595	0.9641
5	0.0000	0.9978	***0.9985***	0.9770	**0.9993**	0.9941
6	0.0000	**0.4776**	0.0066	0.0002	0.0000	***0.1839***
7	0.5671	**0.9430**	***0.8321***	0.0000	0.0000	0.8173
OA	0.2186	0.5123	***0.5705***	0.5513	0.4959	**0.6740**
Kappa	0.1365	0.3907	***0.4355***	0.3630	0.2270	**0.5736**

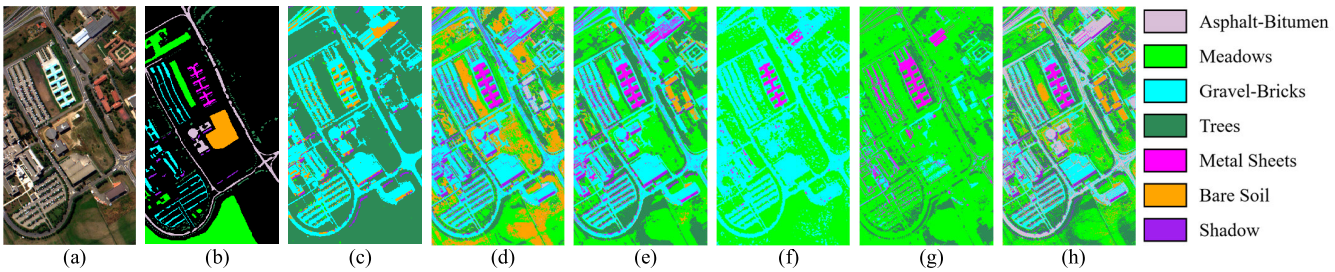


Fig. 14. Classification of different unmixing methods on PaviaU dataset. (a) Pseudo-colored visualization pf PaviaU dataset, (b) reference classification map, (c) ELMM, (d) CNNAEU, (e) MSNet, (f) DeepTrans, (g) MLM-3DAE, and (h) proposed ACR-Net.

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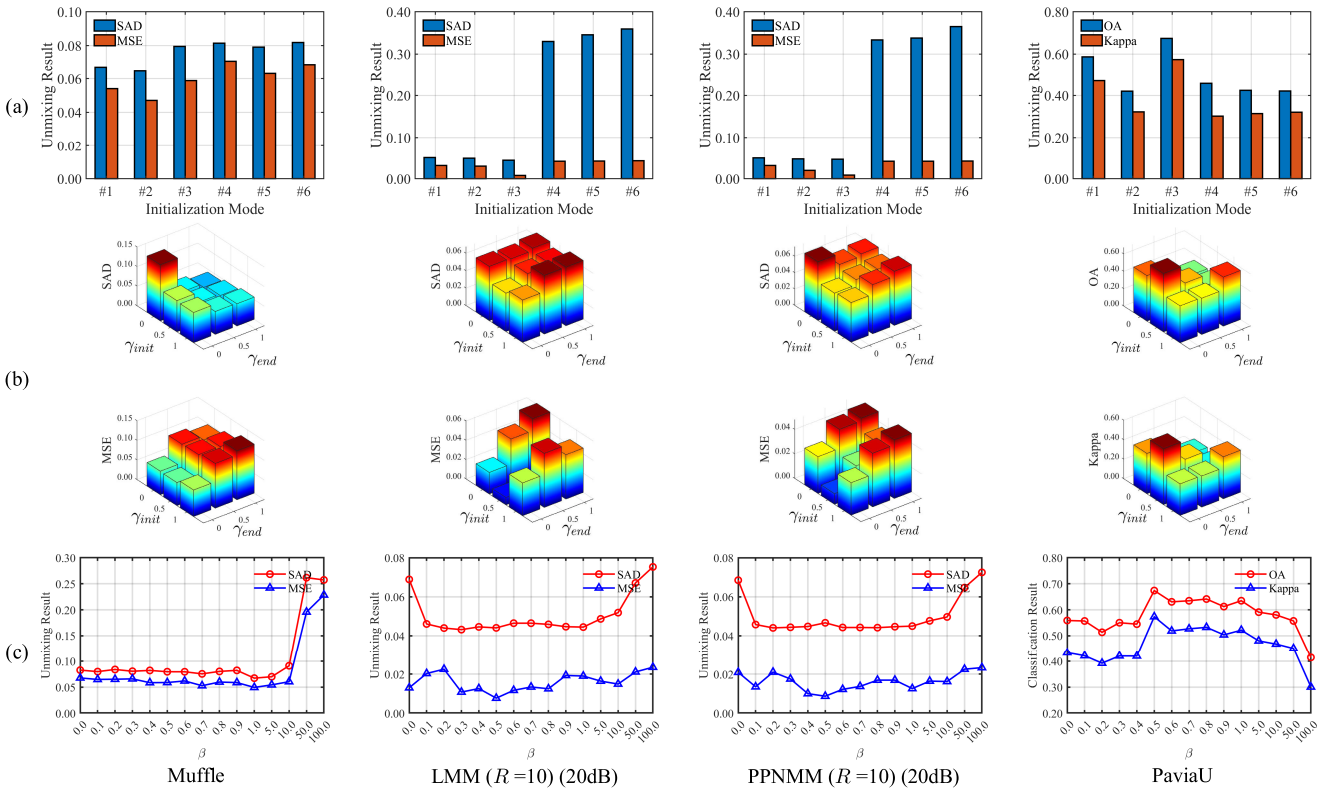


Fig. 15. Impact of (a) network initialization, (b) sparse constraints, and (c) endmember smoothness constraint on unmixing performance for Muffle, LMM ($R = 10$) (20 dB), PPNMM ($R = 10$) (20 dB), and PaviaU dataset.

TABLE VII
INITIALIZATION SETTING FOR INITIALIZATION DISCUSSION

No.	#1	#2	#3	#4	#5	#6
Initialization Mode	Uniform	Normal	He-Normal	Uniform+VCA	Normal+VCA	He-Normal+VCA

[PPNMM ($R = 10$) (20 dB)], and PaviaU dataset. According to Fig. 15(a), the results show that VCA initialization consistently outperforms standard initialization methods across all datasets, particularly in noise-affected synthetic datasets. When the decoder used VCA for initialization, He-normal initialization yielded the best overall performance across all datasets. In conclusion, the combination of predetected endmembers and He-normal initialization is necessary for optimal unmixing performance.

2) Effect of Sparse Constraints on Unmixing: Fig. 15(b) presents the effect of sparse constraints on unmixing performance. The initial and the end value of the coefficient are referred to as γ_{init} and γ_{end}, respectively, where $(\gamma_{\text{init}}, \gamma_{\text{end}})|\gamma_{\text{init}} = [0.0, 0.5, 1.0], \gamma_{\text{end}} = [0.0, 0.5, 0.01]$. The results indicate that unmixing performance is generally poorest at {0.0, 0.0}, {0.0, 1.0}, or {1.0, 1.0}, suggesting that weak sparse constraints are ineffective, while stronger constraints cause binarized abundance, which further degrades performance. Overall, the best unmixing results are achieved under moderate sparse constraint with γ_{init} as 0.5 and γ_{end} as 0.0, where the highest classification accuracy is observed on the PaviaU dataset.

3) Impact of Smoothness Constraints on Unmixing: The experiment of the effect of the endmember smoothness constraint on unmixing performance is shown in Fig. 15(c), with the coefficient β set in the range [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 5.0, 50.0, 100.0]. For the Muffle, LMM ($R = 10$) (20 dB), and PPNMM ($R = 10$) (20 dB) datasets, stable endmembers and narrow fluctuations in abundance performance are observed when β varies between 0.1 and 1.0. Larger β values cause a rapid performance decline. For the PaviaU dataset, classification maps derived from abundance results are evaluated using OA and Kappa due to a lack of unmixing reference. According to Fig. 15(c), the experimental results show that classification performance is poor at beta values of [0.0, 0.1, 0.2, 0.3, 0.5], improves significantly from 0.5 to 1.0, and deteriorates at higher values [5.0, 10.0, 50.0, 100.0]. Overall, $\beta = 0.5$ provides the best comprehensive unmixing and classification performance across all datasets.

4) Effect of CASA and ACAD Module: To test the effectiveness of the CASA module and the ACAD in the proposed ACR-Net, we conducted ablation experiments on various datasets. In these experiments, "Base" represents the base model without either of the two modules, where the decoder contains only a regular convolutional layer. "+CASA" indicates the addition of the CASA module on top of the Base model. Similarly, "+ACAD" signifies the use of ACAD module. Finally, "+CASA&ACAD" indicates the simultaneous incorporation of both designed modules, forming ACR-Net. Table VIII presents the results of the ablation experiments

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TABLE VIII
ABLATION STUDY ON FOUR DATASETS OF ENDMEMBER RESULTS (SAD) AND ABUNDANCE RESULTS (MSE)

Method	Samson		Urban		Muffle	
	mSAD	MSE	mSAD	MSE	mSAD	MSE
Base	0.1994	0.0395	0.1371	0.0181	0.1296	0.0687
+CASA	0.1603	0.0240	0.1276	0.0143	0.0993	0.0596
+ACAD	0.0531	0.0098	0.0722	0.0165	0.0856	0.0683
ACR-Net	0.0335	0.0025	0.0810	0.0149	0.0795	0.0588

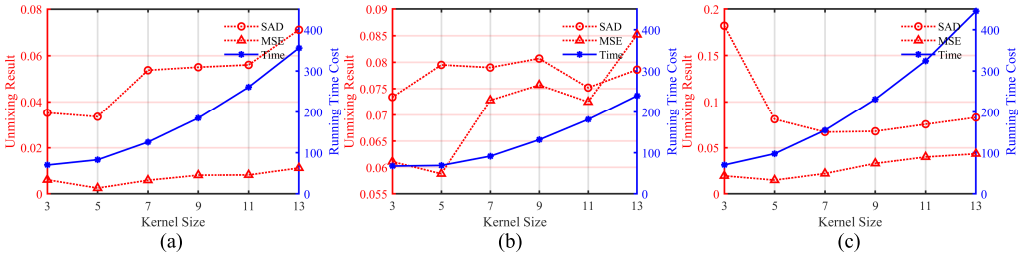


Fig. 16. Effect of kernel size of ACAD on the endmember and abundance performance for (a) Samson dataset, (b) Urban dataset, and (c) Muffle dataset.

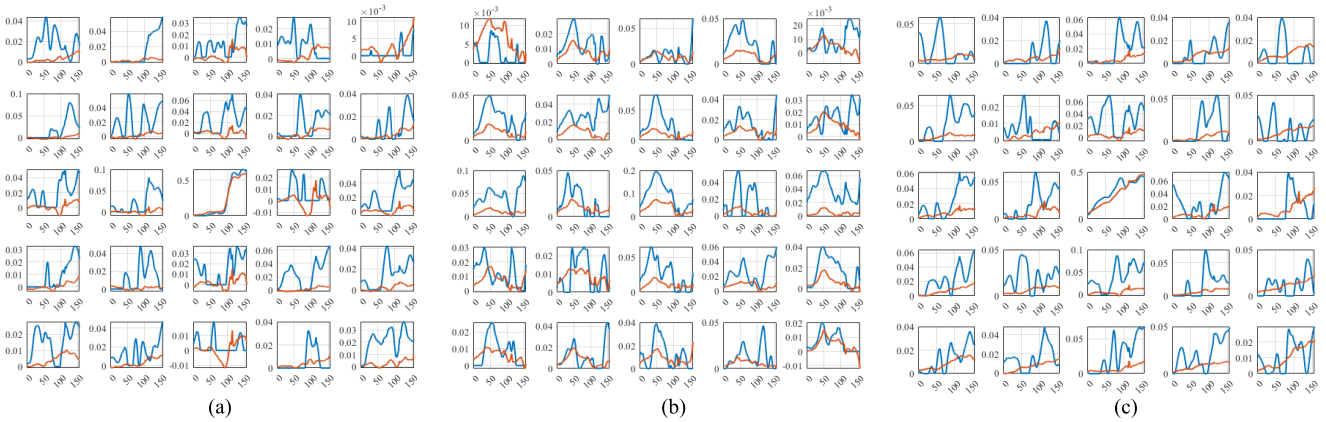


Fig. 17. Visualization comparison of endmembers of the Samson dataset without (in blue) and with (in orange) the ACAD module. (a) Rock, (b) tree, and (c) water.

on the three real datasets. Compared to the Base model, the inclusion of the CASA module results in a certain decrease in both abundance errors for all datasets while also slightly improving endmember extraction accuracy. Adding the ACAD module alone reduces the mSAD value across all datasets, significantly enhancing endmember extraction accuracy. Notably, the inclusion of ACAD leads to a substantial enhancement in endmember performance, while the CASA module is crucial for generating more precise abundance maps. The synergy between these two modules creates a complementary effect, collectively boosting the accuracy of both endmember and abundance estimations.

5) Effect of Convolution Kernel Size in ACAD: Fig. 16 provides an illustration of the effect of convolution kernel size in ACAD on the performance of endmember and abundance, as well as on runtime. Experimental results indicate that a slight increase in kernel size benefits endmember prediction, whereas an overly large kernel size reduces accuracy. Similarly, the pattern of abundance extraction follows a similar trend, and the MSE error initially decreases and then increases. Noted that enlarging the convolution kernel broadens the contextual range for the spectral reconstruction. However, as the distance from the central pixel increases, the number of relevant features may decrease and computational complexity explodes. Therefore, according to the results, a kernel size of 5 emerges as the optimal choice for both endmember and abundance extraction.

6) Visualization of the Influence of ACAD on Endmembers: To further illustrate the impact of ACAD on endmember prediction, we provide a visual comparison of the Samson dataset with and without the ACAD module (presented in Fig. 17). The decoder's convolution kernel size is set as 5×5. In the results of the model without ACAD, a noticeable oscillation in the auxiliary endmembers corresponding to the spatial neighborhood is observed. This phenomenon significantly diminishes when ACAD is introduced. This observation suggests that the incorporation of ACAD leads to more stable endmembers by excluding irrelevant features or noise on the central pixel. This leads to the attainment of more stable and accurate endmembers, ultimately enhancing endmember performance.

G. Running Time Cost

To test the computational efficiency of the ACR-Net, we conducted running time tests on all algorithms (listed in Table IX). The algorithm with the shortest runtime is

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TABLE IX
RUNNING TIME COST COMPARISON FOR FOUR DATASETS

Runnning time	ELMM	CNNAEU	MSNet	DeepTrans	GLU-OH	MLM-3DAE	ACR-Net
Samson	88.7208	88.1974	**19.6403**	***21.4699***	36.7628	187.8733	83.0592
Urban	1897.5997	187.3348	**64.4568**	157.7741	633.4747	2829.8580	***97.8967***
Muffle	190.5890	109.4989	**22.2971**	***28.1000***	46.2914	215.4926	69.2264
LMM (R =10) (20dB)	13368.8793	***113.4648***	**51.0367**	234.8179	4179.3374	2599.8744	338.1212
PPNMM (R =10) (20dB)	85624.3569	***124.9090***	**46.0742**	268.9423	3062.8021	2456.7552	340.6842
PaviaU	2016.9967	***55.3395***	**55.3390**	488.4884	-	4076.2665	690.8767

highlighted in red and bold, while the second best is highlighted in italics and bold. Except for ELMM, which is implemented in MATLAB, the other algorithms are implemented using Python. Due to the large image size of PaviaU dataset, the computational resources required by GLU-OH are too high to run. According to Table IX, MSNet has the shortest runtime across all datasets, while CNNAEU ranks second for all the synthetic datasets and PaviaU dataset. It is observed that ACR-Net's runtime on the three real datasets ranges from 1 to 1.5 min. On datasets with a higher number of endmembers, ACR-Net achieves the highest computational efficiency compared to ELMM, GLU-OH, and MLM-3DAE while delivering comparable unmixing performance and the strongest potential for classification. Overall, ACR-Net demonstrates high computational efficiency with high unmixing performance across various datasets.

IV. CONCLUSION

In summary, this article introduces a ACR-Net for HU. This method establishes a spectral unmixing model with adaptive spatial correlation constraints. Unlike traditional unmixing models with regular spatial information integration, ACR-Net innovatively captures the relevant spatial information for adaptive spectral unmixing, where spatial regions with strong correlations contribute significantly to pixel unmixing while disregarding spatial information unrelated to the pixel. Specifically, a novel CASA module is designed to simultaneously highlight various ground objects, considering the distinctive characteristics of different ground objects across various spectral bands on a spatial scale. Additionally, a novel ACAD mines the spatial correlation from the feature maps of encoder and leverages the affinity map to adjust the contribution of neighbor pixels adaptively, resulting in a more accurate spectral unmixing model. Extensive experimental results confirm the effectiveness and superiority of the proposed method among numerous state-of-the-art algorithms. Furthermore, ablation experiments demonstrate that CASA plays a pivotal role in generating more precise abundance maps, and ACAD significantly enhances endmember performance.

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